

Q1 In this case, a time-dependent variation is observed. Based on the water flow over time, the system is analyzed assuming a dam volume of 270 Km³. The dam's behavior from t=1 to t=9 is modeled and plotted. Although the water level is said to remain constant until July, slight fluctuations are assumed due to climatic and external factors, since perfect stability during that period is physically unrealistic. Then our equation:

$$\frac{dy(t)}{dt} + ay(t) = bu(t) \quad (1.1)$$

$$ay(t) = bu(t) \quad (1.2)$$

$$\frac{dy(t)}{dt} = 0 \quad (1.3)$$

$$Y(s) (s + a) = bU(s)$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b}{s + a} \quad (1.4)$$

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w0=270;
a=[0 2 1.2 3.7 8.9 9.7 14.89 10 15 16];
m=[t=1', 't=2', 't=3', 't=4', 't=5', 't=6', 't=7', 't=8', 't=9', 't=10', 't=11', 't=12'];
v=w0+cumsum(a);
v(10:12)=v(9);
t=1:length(v);
t_fine=linspace(t(1),t(end),200);
v_smooth = interp(t,v,t_fine,'spline');
plot(t_fine,v_smooth,'-b','LineWidth',1.5);
hold on;
plot(t,v,'o','MarkerFaceColor','r');
grid on;
title('Volume of Irrigation Dam by Month');
xlabel('TIME');
ylabel('Amount of the Irrigation Dam (K m³)');
ticks(t);
ticklabels(m);
tickangle(45);
for i=1:length(v)
    text(t(i),v(i)+0.5,num2str(v(i)), 'HorizontalAlignment','center');
end

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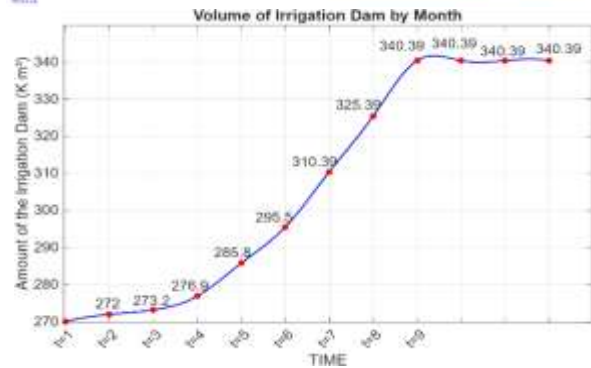


Figure1.1 Volume of Irrigation Dam by Month

As seen in the graph, the system first shows a transient response while filling, then reaches steady-state when $\frac{dy(t)}{dt} = 0$. At this point,

the equation becomes $ay(t) = bu(t)$. and the steady-state gain is $\frac{b}{a}$

Q2 In this part, 8K m³ of water is transferred in July (t=9). This period represents the transient response of the dam. After t=9, the water volume stabilizes at 357.69K m³, indicating that the system has reached the steady-state.

$$y(t) = y_h(t) + \int_0^t h(t-r) du(r) dr \quad (2.1)$$

$$Y(s) = H(s) \cdot U(s) \quad G(s) = \frac{Y(s)}{U(s)}$$

$$G(s) = \frac{H(s) \cdot U(s)}{U(s)} \quad G(s) = H(s) \quad (2.2)$$

y(t): The output y_h(t): The initial effect

t: Present time r: Past time

U(s): Laplace transform of the input u(t)

Y(s): Laplace transform of the output y(t)

```

m=[0 2 1.2 3.7 8.9 9.7 14.89 10 15 16 8 6 3.3 0];
m=[t=1', 't=2', 't=3', 't=4', 't=5', 't=6', 't=7', 't=8', 't=9', 't=10', 't=11', 't=12'];
v=w0+cumsum(a);
t=1:length(v);
t_fine=linspace(t(1),t(end),200);
v_smooth = interp(t,v,t_fine,'spline');
plot(t_fine,v_smooth,'-b','LineWidth',1.5);
hold on;
plot(t,v,'o','MarkerFaceColor','r');
grid on;
title('Volume of Irrigation Dam by Month');
xlabel('TIME');
ylabel('Amount of the Irrigation Dam (K m³)');
ticks(t);
ticklabels(m);
tickangle(45);
for i=1:length(v)
    text(t(i),v(i)+0.5,num2str(v(i)), 'HorizontalAlignment','center');
end

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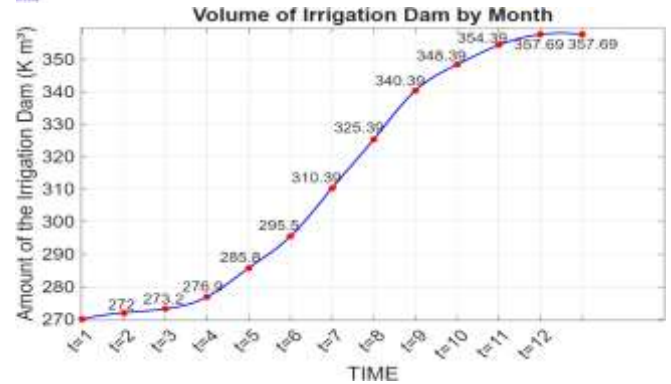


Figure2.1 Volume of Change Irrigation Dam by Month

Q3 In this section, the volume of the irrigation dam gradually increases from t=1 to t=10, reaching 187.69 K m³. The increase between t=4 and t=9 represents the transient period during which the system adapts to input variations. After t=10, the water volume remains constant at 187.69 K m³, indicating that the system has reached a steady-state. This graph illustrates the annual filling trend of the system; therefore, the newly develop model is constructed based on this behavior.

$$\frac{d^2y(t)}{dt^2} + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_0 u(t) \quad (3.1)$$

$$s^2 Y(s) + a_1 s Y(s) + a_0 Y(s) = b_0 U(s)$$

$$Y(s) (s^2 + a_1 s + a_0) = b_0 U(s)$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_0}{s^2 + a_1 s + a_0} \quad (3.2)$$

y(t): System output (dam's volume)

u(t): System input t: Time

a₁, a₀, b₀: Constant coefficients that define the system

s: The complex variable in the Laplace transform

Y(s): Laplace transform of the output y(t)

U(s): Laplace transform of the input u(t)

G(s): The system's Transfer Function (ratio of output to input)

```

V0=100;
x=[0 2 1.2 3.7 8.9 9.7 14.89 15 15 16 8 6 3.3 0];
m=[t=1', 't=2', 't=3', 't=4', 't=5', 't=6', 't=7', 't=8', 't=9', 't=10', 't=11', 't=12'];
v=V0+cumsum(x);
t=1:length(v);
t_fine=linspace(t(1),t(end),200);
v_smooth = interp(t,v,t_fine,'spline');
plot(t_fine,v_smooth,'-b','LineWidth',1.5);
hold on;
plot(t,v,'o','MarkerFaceColor','r');
grid on;
title('Volume of Irrigation Dam by Month');
xlabel('TIME');
ylabel('Amount of the Irrigation Dam (K m³)');
ticks(t);
ticklabels(m);
tickangle(45);
for i=1:length(v)
    text(t(i),v(i)+0.5,num2str(v(i)), 'HorizontalAlignment','center');
end

```

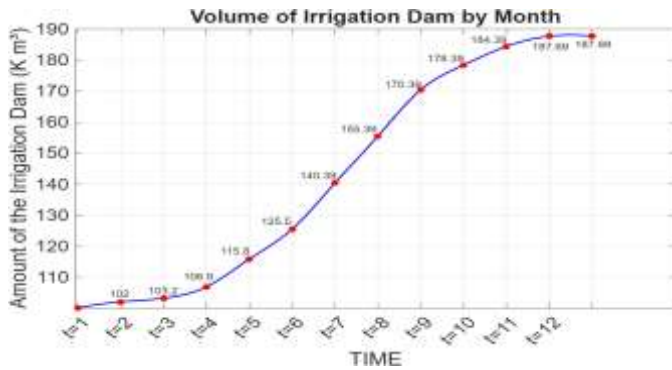


Figure 3.1 Annual Course of Dam Volume

Q4) This graph shows how the dam's water absorption capacity decreases over time. Initially, the absorption is fastest, but as the dam's ground becomes saturated with water, this rate slows down, and when the "saturation" point is reached, the absorption nearly stops. The water absorbed by the dam's ground is the internal uncertainty.

$$y(t) = y(0) e^{-at} \quad (4.1)$$

$$\frac{dy(t)}{dt} = -ay(t) \quad (4.2)$$

$$sY(s) + aY(s) = y(0)$$

$$Y(s)(s + a) = y(0) \quad (4.3)$$

$$Y(s) = \frac{y(0)}{s + a}$$

```
t = 0:0.1:2;
k = 0.8;
y0 = 1000;
y = y0 * exp(-k*t);
plot(t, y, 'b');
hold on;
plot(2, y(end), 'ro');
text(2, y(end)+20, 'Saturation');
xlabel('TIME');
ylabel('Water Absorbed (m³)');
title('Dam Absorption');
xticks([0 1 2]);
xticklabels({'t=1', 't=2', 't=3'});
grid on;
hold off;
```

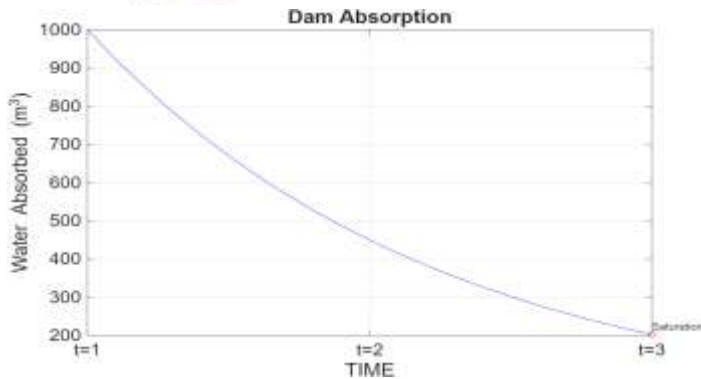
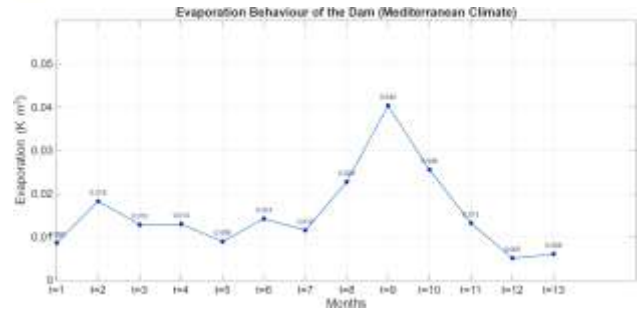


Figure 4.1 Dam Absorption

In October (t=3), it reaches saturation level and the absorption becomes zero.

Q5) It is assumed that evaporation reaches its maximum level in July. However, compared to July, the values in other month especially in the winter months appear to vary randomly. The irrigation dam is assumed to be located in the Mediterranean region, where such a pattern of maximum evaporation is considered typical.

```
months = {'t=1', 't=2', 't=3', 't=4', 't=5', 't=6', 't=7', 't=8', 't=9', 't=10', 't=11', 't=12', 't=13'};
maxE = 0.05;
evap = maxE * (0.6*rand(1,13));
evap(3) = maxE;
evap = smoothdata(evap, 'gaussian', 3);
plot(evap, '-o', 'LineWidth', 1.5, 'MarkerFaceColor', 'b'); grid on;
xticks(1:13); xticklabels(months);
xlabel('Months'); ylabel('Evaporation (K m³)');
title('Evaporation Behaviour of the Dam (Mediterranean Climate)');
ylim([0 maxE*1.2]);
for k = 1:13
    text(k, evap(k)+0.002, sprintf('%1.1f', evap(k)), 'HorizontalAlignment', 'center');
end
```



$$A \frac{dH(t)}{dt} = q_i(t) - q_o(t) - q_{ev}(t) \quad (5.1)$$

$$A[sH(s) - H(0)] = Q_{i(s)} - Q_o(s) - Q_{ev}(s) \quad (5.2)$$

H(t): Measured height of the fluid

q_i : flow rate entering the system q_o : flow rate exiting the system

H(0): the height of the system at the initial moment

Q6) It is assumed that the dam gate independently leaks 0.01 km³ of water. This leakage is an uncertainty and is internal, because it depends on the structure of the irrigation dam itself and is not related to external factors. Therefore, the gate remains closed during rainy months, and the leakage occurs in those months.

$$y_{new} = y_{old} - 0.01 \quad \frac{dy(t)}{dt} = -q_{leak} \quad y(t) = -0.01t + C \quad y(t) = y_0 - 0.01t \quad (6.1)$$

$$\frac{dy(t)}{dt} = q_{in(t)} - q_{out(t)} - q_{leak(t)} \quad (6.2)$$

```
months = {'t=1', 't=2', 't=3', 't=4', 't=5', 't=6', 't=7', 't=8', 't=9', 't=10', 't=11', 't=12', 't=13'};
plot(0:0.01*ones(1,13), '-o'); grid on;
title('Constant Leakage'); xlabel('TIME'); ylabel('Leakage (K m³)');
xticks(1:13); xticklabels(months);
text(7, 0.0105, '0.01', 'HorizontalAlignment', 'center');
```

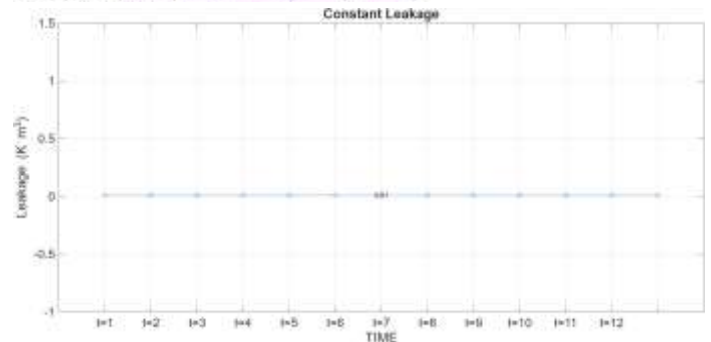


Figure 5.1 Change Constant leakage

Q7) DIAGRAM

